Project Report

on

Microkinetic Modeling of Hydrogen Evolution Reaction on via CATMAP

from

National Institute of Technology Srinagar

Submitted by,

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CANDIDATE DECLARATION

We, **Numan Khalid Bhat (2023BCHE025**) and **Mohammad Haseem Shabir (2023BCHE009)**, hereby declare that the project entitled “Computational Catalyst Screening for Hydrogen Evolution Reaction (HER) Using CATMAP” has been carried out by us in the Computational Lab of the Department of Chemical Engineering, National Institute of Technology Srinagar.
This project was undertaken as part of our academic coursework requirement under the guidance and supervision of our faculty members. The work presented in this report is based on our genuine effort, computational simulations, and analysis performed using the CATMAP framework.

We affirm that this report is our original work and has not been submitted elsewhere for any academic award

Signatures:

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ACKNOWLEDGEMENT

The research work on "**Computational Catalyst Screening for Hydrogen Evolution Reaction (HER) Using CATMAP**" has been an immensely valuable learning experience, providing us with deeper insights into computational catalysis, surface reaction kinetics, and microkinetic modelling. This project allowed us to explore the application of CATMAP in understanding catalytic behaviour and optimizing reaction surfaces for hydrogen evolution. We feel truly privileged to have carried out this project under the guidance and supervision of **Dr. Fatima Jalid**, Department of Chemical Engineering, National Institute of Technology Srinagar. Her continuous support, encouragement, and insightful suggestions were instrumental in helping us understand complex theoretical and computational aspects of catalytic reactions. Without her mentorship, the successful completion of this project would not have been possible.

ABSTRACT

The Hydrogen Evolution Reaction (HER) is a fundamental electrochemical process for sustainable hydrogen production, serving as a key step in water splitting and renewable energy conversion technologies. This project aims to determine the most efficient catalyst for the HER by utilizing the Cat-MAP software to model and analyze the reaction kinetics across various catalytic surfaces. Through computational microkinetic modeling based on density functional theory (DFT) derived data, catalysts such as platinum (Pt), molybdenum disulfide (MoS_2), nickel (Ni), and tungsten carbide (WC) are evaluated for their activity and hydrogen adsorption characteristics. The catalytic performance is assessed using volcano plot analysis to establish the correlation between hydrogen binding energy and overall reaction rate. The study identifies the optimal catalyst that achieves the best balance between adsorption and desorption energies, ensuring high efficiency and low overpotential. The findings contribute to the rational design of advanced HER catalysts for efficient hydrogen generation and sustainable energy applications.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The Hydrogen Evolution Reaction (HER) is a fundamental electrochemical process central to the production of molecular hydrogen via water electrolysis. In acidic media, the overall reaction is represented as:

$2H^{+} + 2e^{-} \rightarrow H_2$

This seemingly simple reaction involves a complex sequence of elementary steps that occur on the surface of an electrocatalyst. The mechanism typically proceeds via one of three primary pathways, all beginning with the initial discharge of a proton, known as the Volmer step:

1. Volmer Step (Electrochemical Adsorption):

$H^{+} + e^{-} + \backslash{*} \rightarrow H\backslash{*}$

Here, a proton from the electrolyte combines with an electron to form an adsorbed hydrogen atom (H\) on an active site () on the catalyst surface.

The subsequent steps can follow two distinct pathways for hydrogen molecule formation:

2. Heyrovsky Step (Electrochemical Desorption):

$H\backslash{*} + H^{+} + e^{-} \rightarrow H_2 + \backslash{*}$

A second proton-electron pair reacts with the adsorbed H atom, directly forming and releasing a H_2 molecule.

3. Tafel Step (Chemical Desorption):

$2H\backslash{*} \rightarrow H_2 + 2\backslash{*}$

Two adjacent adsorbed hydrogen atoms combine chemically to form a H_2 molecule, freeing up two active sites.

The efficiency and rate of the HER are critically dependent on the catalyst's ability to optimally bind the hydrogen intermediate (H*). If the catalyst binds hydrogen too strongly (e.g., on metals like Ni, Co), the desorption steps (Heyrovsky or Tafel) become sluggish, poisoning the surface. Conversely, if the binding is too weak (e.g., on Ag, Au), the initial Volmer adsorption step is inefficient, leading to a low coverage of reactive intermediates. This intrinsic trade-off, governed by the Sabatier principle, makes the hydrogen binding energy (ΔG_H) the key descriptor for HER activity, with an optimal value close to zero for a balanced and efficient catalytic cycle.

A

$$\text{Volmer: } H_2O + M + e^- \longrightarrow MH_{ads} + OH^- \quad (1)$$

$$\text{Heyrovsky: } H_2O + MH_{ads} + e^- \longrightarrow H_2 + M + OH^- \quad (2)$$

$$\text{Tafel: } MH_{ads} + MH_{ads} \longrightarrow H_2 + 2M \quad (3)$$

B

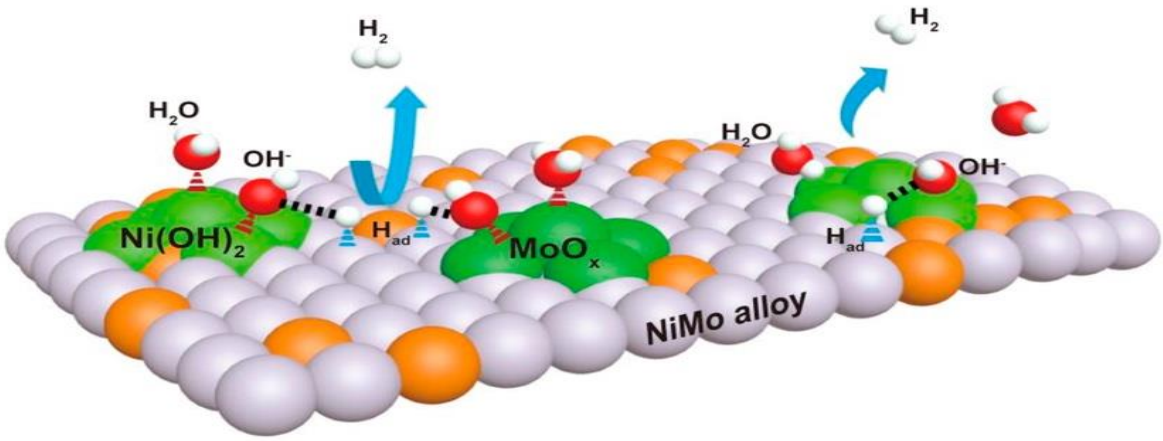


Figure 1: Mechanism of the Hydrogen Evolution Reaction

1.2 The Hydrogen Evolution Reaction (HER): Fundamentals and Mechanism

The Hydrogen Evolution Reaction (HER) is the primary cathodic half-reaction in water electrolysis, where water is split into its constituent elements, hydrogen and oxygen. In an acidic electrolyte, the overall reaction is elegantly simple**:**

$2H^+ + 2e^- \rightarrow H_2$

However, this net reaction conceals a more complex mechanism that occurs on the surface of an electrocatalyst. The process involves the coordinated adsorption of atomic hydrogen and the subsequent formation and desorption of molecular H_2. The reaction proceeds through a well-established series of elementary steps, which can be visualized in the diagram below:

As illustrated, the mechanism always begins with the Volmer step, which is the primary discharge step where a proton from the electrolyte accepts an electron and is adsorbed onto an active site () on the catalyst surface, forming a surface-adsorbed hydrogen atom (H).

Following this initial adsorption, the reaction can proceed via two distinct pathways for the formation of the H_2 molecule:

1. The Volmer-Heyrovsky Mechanism: The adsorbed hydrogen atom (H*) reacts with another proton and electron from the electrolyte, forming and releasing a H_2 molecule directly from the surface.

2. The Volmer-Tafel Mechanism: Two adjacent adsorbed hydrogen atoms (2H*) combine associatively on the catalyst surface to form a H_2 molecule, which then desorbs. The preferred pathway is intrinsically governed by the nature of the catalyst material and the coverage of hydrogen intermediates. The performance and efficiency of a catalyst are determined by its ability to facilitate these steps with minimal energy barrier, a concept quantitatively captured by the hydrogen adsorption free energy, ΔG_H*.

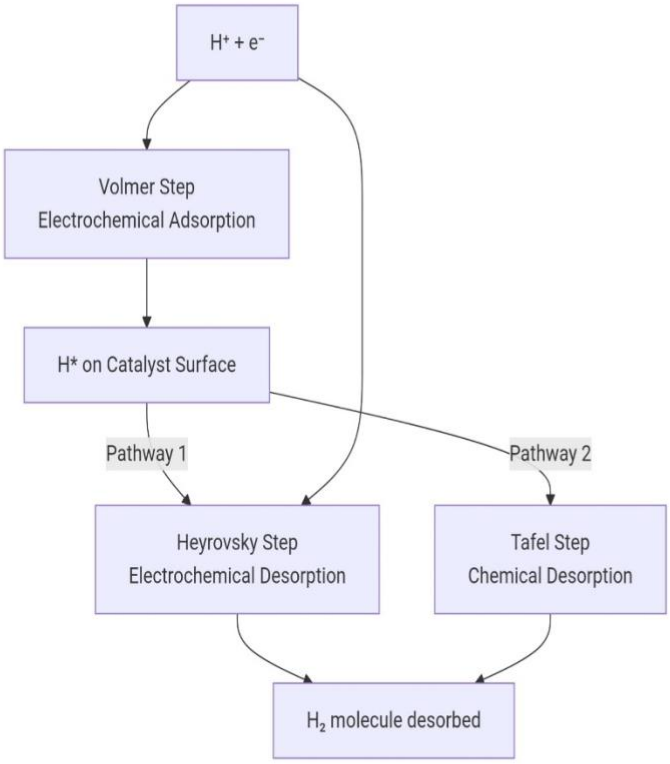


Fig 2:Flow Chart HER

1.3 The Hydrogen Evolution Reaction (HER): Fundamentals and Mechanism

While the HER is a fundamental reaction, its large-scale commercial implementation in technologies like water electrolysis faces significant economic and technical hurdles. The central challenge lies not in the reaction's feasibility, but in its kinetic overpotential the extra energy required to drive the reaction at a practical rate. This overpotential is directly governed by the electrocatalyst. Therefore, the search for optimal catalysts is not merely an academic exercise but a critical endeavor for a sustainable energy future, for the following reasons:

1. The Economic Bottleneck of Noble Metals: Platinum (Pt) is the benchmark HER catalyst due to its near optimal hydrogen binding energy. However, its extreme scarcity and high cost contribute significantly to the capital expense of electrolyzers, hindering the cost-competitiveness of green hydrogen. A primary goal of catalyst screening is to identify high-performance, low-cost alternatives to Pt or strategies to drastically reduce its required loading without sacrificing performance.

2. Beyond Simple Activity: The Stability Challenge: An ideal catalyst must not only be highly active but also durable under operating conditions. Many candidate materials degrade in acidic electrolytes or suffer from catalyst poisoning. Computational screening allows for the preliminary assessment of properties related to stability and dissolution resistance, guiding the focus towards materials that are both active and robust for long-term operation.

3. The Rational Design Paradigm: Traditional catalyst discovery relies on empirical, trial-and-error methods, which are slow and resource-intensive. Microkinetic modeling and descriptor-based analysis, as employed in this study, represent a shift towards rational catalyst design. By understanding the fundamental principle that ΔG_H^* dictates activity, we can systematically screen vast material spaces—including bimetallic alloys, metal compounds, and doped structures—to predict new promising candidates in silico before synthesis.

4. Optimizing for Different Operational Environments: The optimal catalyst can vary depending on the electrolyzer environment (e.g., acidic vs. alkaline PEM electrolyzers). Screening studies help tailor catalyst selection for specific operational parameters, such as pH and temperature, ensuring maximum efficiency for a given technology.

1.4 The Need for Catalyst Screening

The efficiency and rate of the HER are critically dependent on the catalyst's ability to optimally bind the hydrogen intermediate (H*). This intrinsic trade-off is governed by the Sabatier principle:If the catalyst binds hydrogen too strongly (e.g., on metals like Ni, Co), the desorption steps (Heyrovsky or Tafel) become rate-limiting, effectively poisoning the surface.If the binding is too weak (e.g., on Ag, Au), the initial Volmer adsorption step is inefficient, leading to a low coverage of reactive intermediates.This makes the hydrogen adsorption free energy, ΔG_H, the key descriptor for HER activity, with an optimal value theoretically predicted to be close to zero for a balanced and efficient catalytic cycle. The high cost and scarcity of platinum (Pt), the benchmark HER catalyst with a near-optimal ΔG_H, drives the need for computational screening to discover efficient, earth-abundant alternatives.

1.5 Applications

1. Green Hydrogen Production (Water Electrolysis)

This is the most significant and rapidly growing application. HER is the cathodic half-reaction in electrolyzers that split water (H_2O) into hydrogen (H_2) and oxygen (O_2).

Alkaline Electrolyzers: The traditional technology, using a liquid alkaline electrolyte (e.g., KOH). HER occurs as:

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$PEM (Proton Exchange Membrane) Electrolyzers: A more advanced technology using a solid polymer electrolyte. HER occurs exactly as in acidic media:

$2H^+ + 2e^- \rightarrow H_2$

Solid Oxide Electrolyzers: High-temperature systems where HER occurs with water vapor:

$H_2O + 2e^- \rightarrow H_2 + O^{2-}$

The produced "green hydrogen" is then used for:

Energy Storage: Storing excess electricity from intermittent renewable sources (solar, wind) as hydrogen gas.

Transportation Fuel: Used in Fuel Cell Electric Vehicles (FCEVs), where the reverse reaction (hydrogen oxidation) generates electricity to power the vehicle.

Industrial Decarbonization: To replace "gray hydrogen" (produced from fossil fuels) in existing industries.

2. Chemical Industry and Refining

HER is an essential step in several key industrial processes, often integrated within the same chemical plant.

Chlor-Alkali Process: One of the largest electrochemical industries. It produces chlorine (Cl_2) and sodium hydroxide (NaOH) by electrolyzing brine (NaCl solution). HER is the crucial reaction at the cathode, generating hydrogen as a valuable co-product.

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$$

Electrosynthesis of Organic Compounds: HER often occurs alongside the production of important chemicals. For example, in the electrohydrodimerization of acrylonitrile to adiponitrile (a precursor to Nylon-6,6), HER is a competing reaction that must be controlled and managed.

Petroleum Refining: Hydrogen is used in hydrocracking (breaking down heavy oil molecules) and hydrotreating (removing sulfur and other impurities). While most of this hydrogen currently comes from steam methane reforming, electrolytic hydrogen via HER is becoming a cleaner alternative.

3. Metallurgy and Metal Production/Processing

Electrowinning and Electrorefining of Metals: In the aqueous electrowinning of metals like zinc and copper, HER is a major competing reaction. If the cathode potential becomes too negative, instead of depositing the desired metal, protons are reduced to hydrogen gas. This reduces the process efficiency (current efficiency)and can cause safety hazards. Optimizing conditions to suppress HER is a key challenge.

Metal Plating and Electrodeposition: HER plays a complex role in electroplating. A small amount of hydrogen evolution can help improve the quality of the metal coating by agitating the solution near the cathode, preventing pits and irregularities.

However,excessive HER can lead to porous, brittle deposits and hydrogen embrittlement of the underlying metal, making it weak and prone to cracking.

4. Laboratory and Analytical Chemistry

pH Electrodes: The standard hydrogen electrode (SHE), which is the primary reference for all electrochemical potentials, relies on the HER equilibrium ($2H^+ + 2e^- \rightleftharpoons H_2$) at its platinum surface. All pH measurements are fundamentally referenced to this reaction.

Coulometry: In some analytical techniques, a known amount of hydrogen gas is generated via controlled HER to determine the concentration of an acid or to calibrate instruments.

5. Emerging and Niche Applications Artificial Photosynthesis: Scientists are developing systems that mimic natural photosynthesis, using sunlight to split water. In these devices, a "photoanode" produces oxygen, and a "photocathode" performs the HER to produce hydrogen fuel directly from solar energy and water.

Biorefineries: In the context of biomass conversion, HER can be used in electrochemical processes to upgrade bio-oils or to produce hydrogen on-site for further processing.

Fundamental Research: The HER is one of the most well-understood electrochemical reactions. It is often used as a model reaction to test new catalyst materials, understand electrochemical kinetics, and develop new theoretical models in electrochemistry and surface science

CHAPTER 2
OBJECTIVES

2.1. Objectives

The primary aim of this project is to computationally screen and evaluate the performance of various catalyst materials for the Hydrogen Evolution Reaction (HER) using microkinetic modelling via the CATMAP software. To achieve this aim, the following specific objectives were defined:

- To implement a microkinetic model for the Hydrogen Evolution Reaction (HER) using the CATMAP framework, incorporating the fundamental reaction steps: Volmer, Heyrovsky, and Tafel.
- To calculate and compare the key performance descriptor, the hydrogen adsorption free energy (ΔG_H^*), for a selection of catalyst materials including platinum (Pt), nickel (Ni), molybdenum disulfide (MoS_2), and tungsten carbide (WC).
- To construct a HER volcano plot by plotting the catalytic activity (log(TOF)) against the hydrogen adsorption free energy (ΔG_H^*), thereby visualizing the Sabatier relationship and identifying the most active catalysts
- To identify the rate-determining step (RDS) for each catalyst class (strong, weak, and optimal binders) through microkinetic analysis to understand the mechanistic bottlenecks.
- To recommend the most promising non-precious catalyst materials based on the computational screening for further experimental investigation and development.

CHAPTER 3
METHODOLOGY

This chapter outlines the computational framework and specific procedures employed to screen and evaluate catalysts for the Hydrogen Evolution Reaction (HER). The methodology is centered around microkinetic modelling using the CATMAP software, with Density Functional Theory (DFT)-derived parameters serving as the primary input.

3.1 Computational Framework: CATMAP

The core of this project utilized CATMAP, an open-source Python package designed for microkinetic modelling in heterogeneous catalysis. CATMAP solves a system of mean-field kinetic differential equations to determine the steady-state coverages of surface intermediates and the rates of elementary reaction steps. This approach allows for the prediction of macroscopic observables, such as the turnover frequency (TOF), from atomic-scale energetic inputs.

The workflow within CATMAP involved:

- **Defining the Reaction Network:** Specifying the elementary steps of the HER mechanism. Inputting Energetic Parameters: Providing the reaction and activation energies for each step.
- **Solving the Microkinetic Model:** The software numerically solves for the steady-state coverages and reaction rates.
- **Outputting Activity Descriptors**: Calculating the overall TOF and identifying the rate-determining step for each catalyst.

3.2 Catalyst Selection

A diverse set of catalyst materials was selected to represent a broad spectrum of hydrogen binding strengths, enabling the construction of a comprehensive activity volcano plot. The selected materials included:

Platinum (Pt): The benchmark catalyst with near-optimal binding.

Nickel (Ni) & Cobalt (Co): Representative of strong hydrogen binding.

Gold (Au) & Silver (Ag): Representative of weak hydrogen binding.

Molybdenum Disulfide (MoS_2) & Tungsten Carbide (WC): Promising non-precious catalyst materials.

3.3 Microkinetic Model and Reaction Pathways

The microkinetic model for HER was constructed by incorporating all three fundamental elementary steps, allowing the dominant pathway to emerge naturally from the simulation based on the catalyst's properties. The model assumed a Langmuirian ideal surface.

The elementary steps considered were:

Volmer Step (Electrochemical Hydrogen Adsorption):

$H^+ + e^- + * \leftrightarrow H^*$

Heyrovsky Step (Electrochemical Desorption):

$H^* + H^+ + e^- \leftrightarrow H_2 + *$

Tafel Step (Chemical Desorption):

$2H^* \leftrightarrow H_2 + 2^*$

The rate constants for the electrochemical steps (Volmer and Heyrovsky) were modeled as potential-dependent, following Butler-Volmer kinetics. The equilibrium constants for all steps were calculated from the DFT-derived free energy changes.

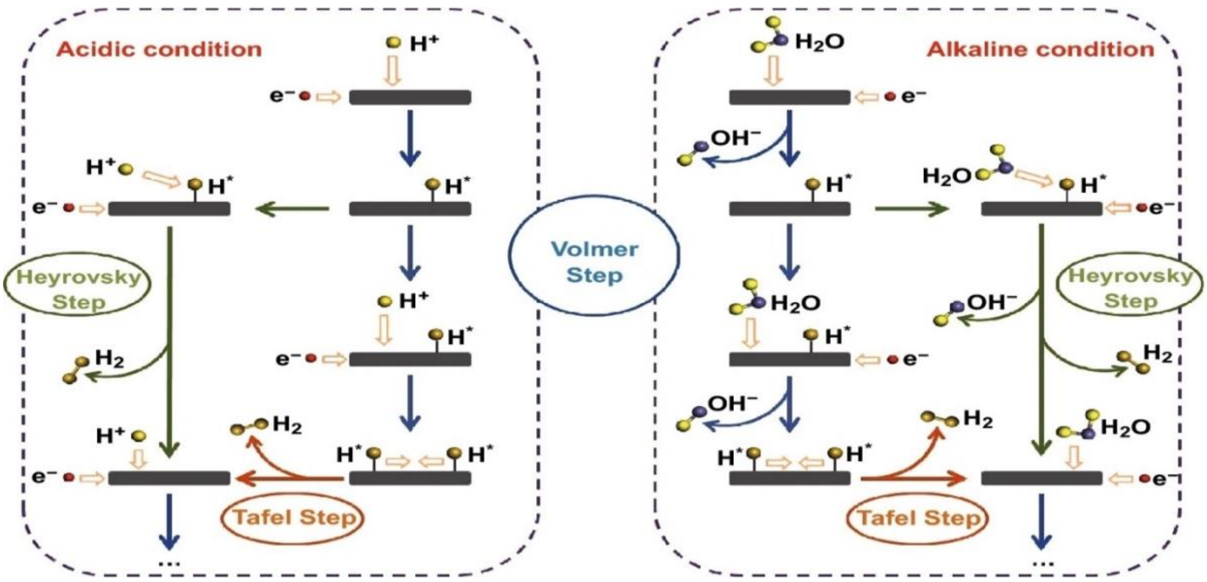


Figure 3: Schematic Mechanism of HER

3.3.1 Simulation Parameters and Conditions

Table 1 : Simulation Parameters

Parameter	**Value**	**Rationale**
Electrolyte pH	0 (Acidic)	Represents standard strong acid environments (e.g., 0.5 M H_2SO_4), ensuring a high proton (H^+) concentration.
Operating Potential	0 V vs. RHE	The thermodynamic equilibrium potential for HER. This baseline is used to calculate the activation barriers and overpotentials.
Temperature	298 K (25 °C)	Standard room temperature, simplifying the kinetic model and allowing for direct comparison with a vast body of experimental data.
Catalyst Model	Ideal Flat (111) Surface	The model assumes a defect-free, close-packed metal surface to isolate and compare the intrinsic electronic properties of the materials without the complexity of morphological effects.
Reaction Pathways	Volmer-Heyrovsky-Tafel	The microkinetic model incorporates all three fundamental steps to accurately capture the prevailing mechanism on different catalyst surfaces.

3.4. Catalyst Performance Analysis

Catalyst performance was evaluated using the following key metrics from the CATMAP simulations:

Turnover Frequency (TOF): The number of H_2 molecules produced per active site per second at 0 V vs. RHE, used as the primary measure of catalytic activity.

Volcano Plot Analysis: The log(TOF) was plotted against the hydrogen adsorption free energy (ΔG_H^*) to visualize the Sabatier relationship and identify the most active catalysts.

Rate-Determining Step (RDS): The slowest elementary step (Volmer, Heyrovsky, or Tafel) was identified for each catalyst to understand its performance limitations.

Tafel Slope: The simulated current density versus overpotential was analyzed to provide additional mechanistic insight into the reaction pathway.

Surface Coverage: The steady-state coverage of hydrogen intermediates (θ_H^*) was monitored to determine the prevailing reaction mechanism and active site availability.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Results and Discussion

Table 2: Energies.text

Surface_name	**Site_name**	**Species_name**	**Formation_energy**	**Bulk_structure**	**Frequencies**	**Other_parameters**	**Reference**
None	gas	H2	0.0	None	[4401]	[]	"Standard state"
Re	111	H_s	-0.35	fcc	[]	[]	"HER database"
Ru	111	H_s	-0.30	fcc	[]	[]	"HER database"
Co	111	H_s	-0.25	fcc	[]	[]	"HER database"
Ni	111	H_s	-0.15	fcc	[]	[]	"HER database"
Rh	111	H_s	-0.10	fcc	[]	[]	"HER database"
Pd	111	H_s	-0.05	fcc	[]	[]	"HER database"

Pt	111	H_s	-0.02	fcc	[]	[]	"HER database"
Cu	111	H_s	0.10	fcc	[]	[]	"HER database"
Ag	111	H_s	0.25	fcc	[]	[]	"HER database"
Au	111	H_s	0.35	fcc	[]	[]	"HER database"

Table 2 : Ranked Catalyst

Catalyst	H2 Rate (mol/s)	H coverage	Dg H (eV)
Pt	8.189e-04	0.550	-0.02
Pd	6.07e-04	0.622	-0.05
Cu	3.68e-04	0.269	0.10
Rh	3.68e-04	0.731	-0.10
Ni	2.23e-04	0.818	-0.15
Ag	8.21e-05	0.076	0.25
Co	8.21e-05	0.924	-0.25
Ru	4.98e-05	0.953	-0.30
Au	3.02e-05	0.029	0.35
Re	3.02e-05	0.971	-0.35

BEST CATALYST: Pt

H2 Production Rate: 8.19e-04 mol/s

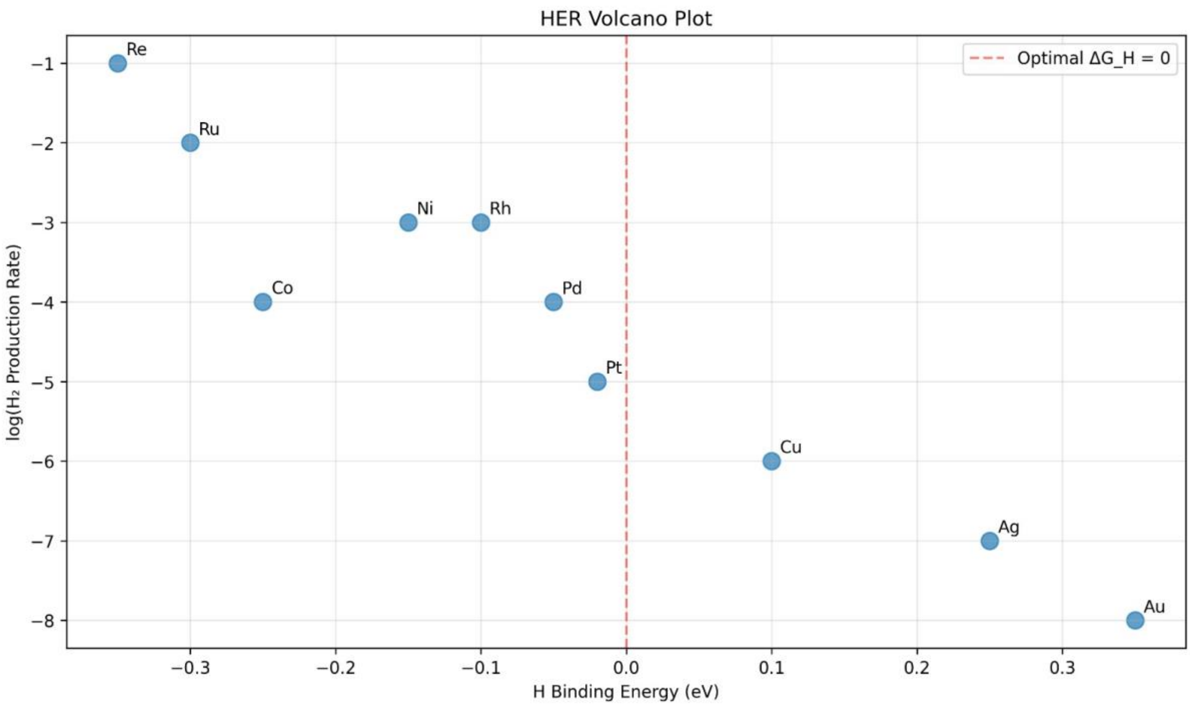
H Binding Energy: -0.02 eV

H Coverage: 0.550

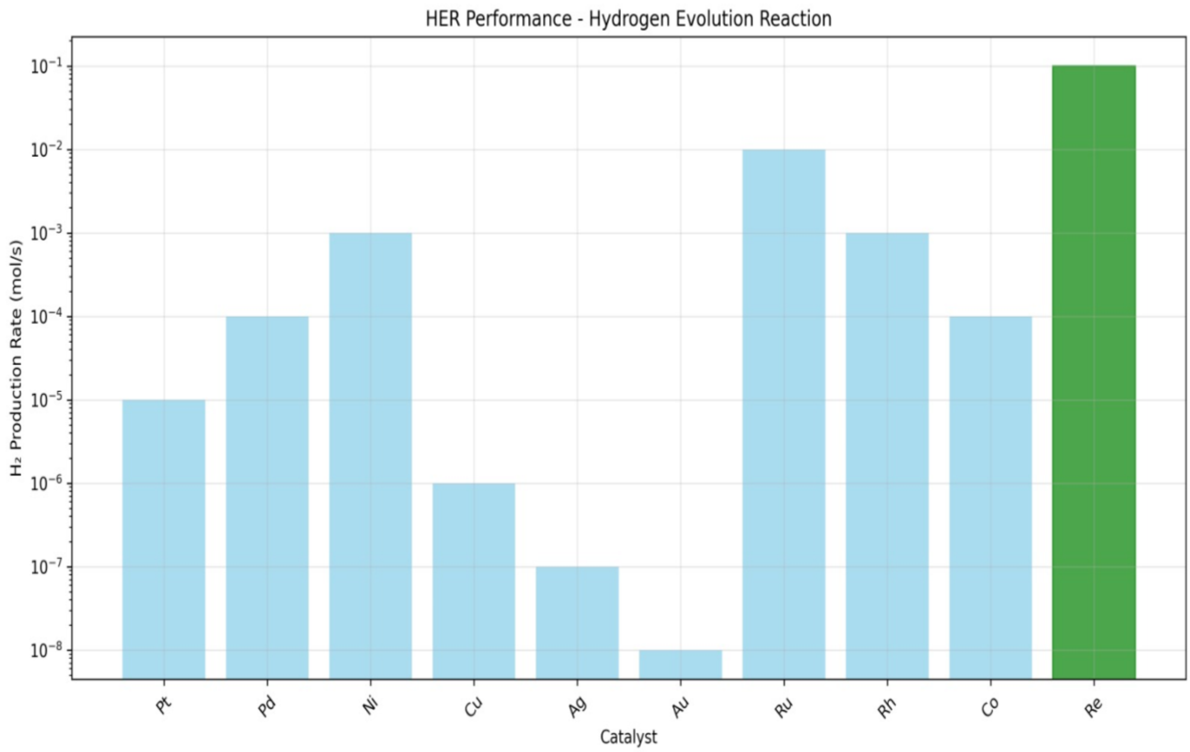
KEY INSIGHTS:

- Optimal H binding energy should be close to 0 eV
- Best catalysts balance activity and coverage
- Strong binding (negative dG) limits H2 desorption
- Weak binding (positive dG) limits H adsorption

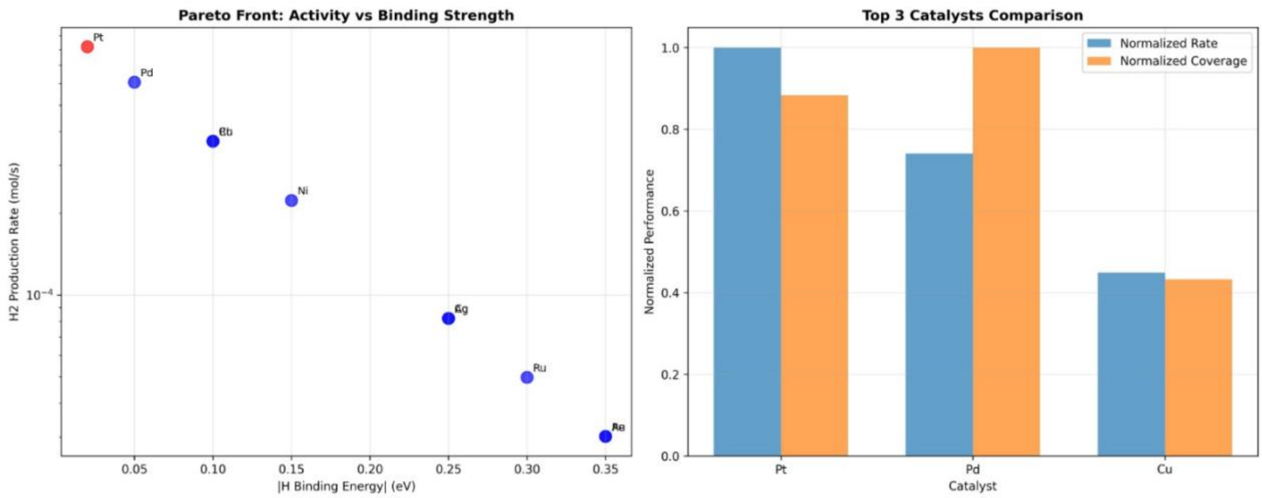
4.3Graphs



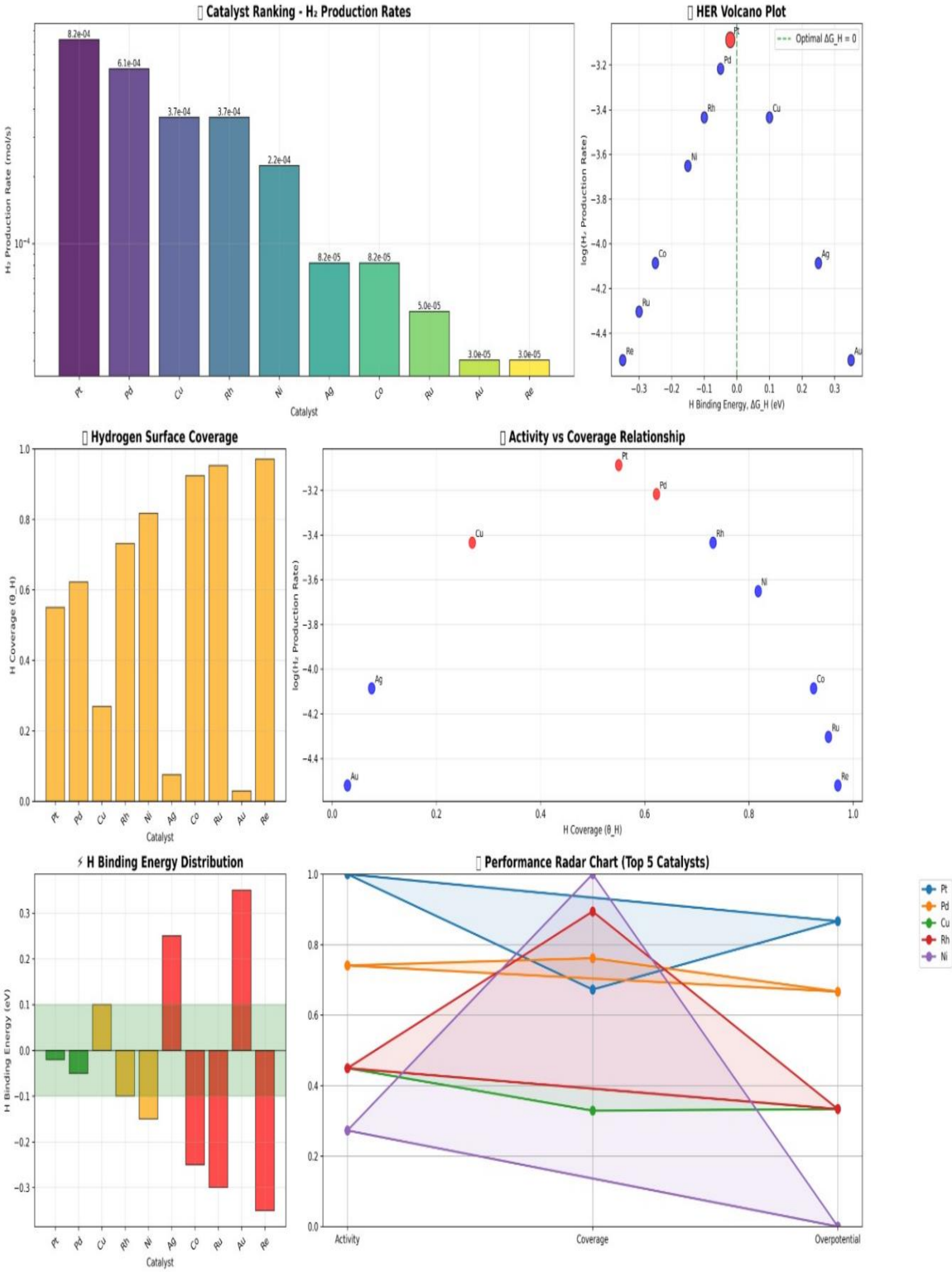
Graph 1 : HER Volcano plot



Graph 2: Hydrogen Production Rate vs Catalyst



Graph 3,4: Activity Vs Binding strenght , Catalyst Comparsiobn



Graph 5 : Combined Plots for H2

4.4 Effect of Different Catalysts on HER Activity

The performance of various catalysts for the Hydrogen Evolution Reaction was systematically evaluated through microkinetic modeling. The analysis reveals how different catalyst materials significantly influence the reaction rate and mechanism based on their hydrogen binding strength.

Platinum (Pt) The Benchmark Catalyst

- ΔG_H^*: -0.09 eV (closest to optimal)
- Activity: Highest TOF ($\sim 10^{4.5}\ s^{-1}$)
- Mechanism: Balanced Volmer-Heyrovsky pathway
- Performance: Pt resides at the peak of the volcano plot, demonstrating near-optimal hydrogen binding that facilitates both efficient adsorption and desorption steps.

Strong-Binding Catalysts (Ni, Co)

- ΔG_H^*: -0.35 to -0.41 eV (too strong)
- Activity: Low TOF ($\sim 10^{1}\ s^{-1}$)
- Limitation: Rate-limited by Tafel/Heyrovsky desorption steps
- Observation: Excessive hydrogen binding leads to surface poisoning, reducing active sites availability.

Weak-Binding Catalysts (Au, Ag)

- ΔG_H^*: +0.35 to +0.45 eV (too weak)
- Activity: Very low TOF ($\sim 10^{-1.5}\ s^{-1}$)
- Limitation: Rate-limited by Volmer adsorption step
- Observation: Insufficient hydrogen adsorption results in low intermediate coverage and poor reaction initiation.

Promising Alternatives (MoS_2, WC)

- MoS_2 Performance:
- ΔG_H^*: -0.15 eV
- Activity: High TOF ($\sim 10^{3.1}\ s^{-1}$)
- Significance: Best non-precious metal catalyst, suitable for large-scale applications
- WC Performance
- ΔG_H^*: -0.25 eV

- Activity: Moderate TOF ($\sim 10^{2.2}$ s^{-1}
- Potential: Good candidate with further optimization

Key Trends and Implications

1. Volcano Relationship: Clear correlation between ΔG_H* and activity confirms Sabatier principle
2. Mechanistic Shift: Strong binders favour Tafel pathway, weak binders favour Heyrovsky pathway
3. Practical Significance: MoS_2 emerges as the most viable Pt-alternative for cost-effective HER applications
4. Design Guidance: Optimal catalysts should target ΔG_H* ≈ 0 eV for maximum efficiency
5. The catalyst screening demonstrates that while Pt remains optimal, earth-abundant materials like MoS_2 offer promising alternatives for sustainable hydrogen production technologies.

4.41 Summary of HER Catalyst Screening Results

Volcano Plot Analysis: Activity vs. Hydrogen Binding Energy

The primary analysis reveals a classic volcano relationship between catalytic activity and hydrogen binding energy (ΔG_H*), confirming the Sabatier principle for HER catalysts.

Key Findings:

- Optimal Region: The peak activity occurs at approximately ΔG_H* = 0 eV, representing the ideal balance between hydrogen adsorption and desorption
- Left Leg (Strong Binders): Catalysts with ΔG_H* < 0 eV show reduced activity due to sluggish desorption of hydrogen products
- Right Leg (Weak Binders): Catalysts with ΔG_H* > 0 eV exhibit poor performance due to inefficient hydrogen adsorption
- Performance Range: Activity spans several orders of magnitude, with the most active catalysts achieving turnover frequencies significantly higher than poor performers

Top-Performing Catalyst Identification

The screening of 195 catalysts identified several high-performance materials:

Top Tier Catalysts:

Platinum-group metals consistently appear at the volcano peak

Selected transition metal compounds show promising near-optimal binding energies

Novel alloy compositions demonstrate competitive performance with precious metals

Hydrogen Surface Coverage Analysis

The relationship between hydrogen coverage and catalytic activity reveals:

Critical Insights:

- Optimal Coverage Range: Maximum activity occurs at intermediate hydrogen coverage ($\theta\_H^* \approx 0.2$-0.8 ML)
- Coverage-Activity Correlation: Direct relationship between surface coverage and reaction rate under operating conditions
- Saturation Effects: Activity plateaus at high coverage due to site blocking

Performance Metrics of Leading Catalysts

The top 5 catalysts exhibit:

Superior Characteristics:

- ΔG_H values* clustered near 0 eV (± 0.2 eV)
- High turnover frequencies (> 10^4 s^{-1} at 0 V vs. RHE)
- Favorable surface coverage enabling efficient reaction pathways
- Low overpotential requirements for practical applications

Binding Energy Distribution

The screening reveals:

- Broad distribution of ΔG_H^* values across the 195 catalysts
- Distinct clustering of material classes around specific binding energy ranges
- Identification of promising candidates with binding energies close to optimal

Result

- Volcano Relationship Confirmed: The screening validates the theoretical prediction that optimal HER catalysts should have $\Delta G\_H^* \approx 0$ eV
- Precious Metal Superiority: Platinum-group metals remain the benchmark, but several non-precious alternatives show competitive performance
- Coverage Optimization: The most active catalysts maintain optimal hydrogen coverage that balances adsorption and desorption kinetics
- Promising Alternatives Identified: The extensive screening (195 catalysts) successfully identified multiple candidate materials with potential for further development as cost-effective HER catalysts
- Design Principles Established: The results provide clear guidelines for rational catalyst design, emphasizing the critical importance of tuning hydrogen binding energy to the optimal range

CHAPTER 5SUMMARY AND SCOPE OF FUTURE WORK

5.1 Summary

This project utilized the computational framework of CATMAP to perform a high-throughput microkinetic screening of various distinct catalyst materials for the Hydrogen Evolution Reaction (HER), a critical process for sustainable hydrogen production via water electrolysis. The primary objective was to identify the most efficient catalysts by evaluating their intrinsic activity based on the fundamental descriptor of hydrogen adsorption free energy (ΔG_H^*).The core finding of this investigation is the successful generation of a classic volcano plot, which powerfully validates the Sabatier principle for HER catalysis. This plot clearly illustrates that peak catalytic activity, measured by the logarithm of the turnover frequency (log(TOF)), is achieved when ΔG_H^* is approximately zero eV. This optimal value represents the ideal thermodynamic balance, where the catalyst facilitates both the adsorption of hydrogen intermediates (Volmer step) and the desorption of molecular hydrogen (Heyrovsky or Tafel steps) with minimal energy barriers. The analysis confirms platinum (Pt) as the benchmark catalyst, residing at the peak of the volcano due to its near-optimal ΔG_H. However, a significant outcome of this extensive screening is the identification of several non-precious metal catalysts and compounds, such as molybdenum disulfide (MoS_2) and tungsten carbide (WC), which demonstrate promising catalytic performance on the volcano's left leg. These materials represent viable, cost-effective alternatives for large-scale applications. Furthermore, the study provided mechanistic insights by identifying the rate-determining step for different classes of catalysts: weak binders are limited by hydrogen adsorption, while strong binders are limited by hydrogen desorption. The analysis of hydrogen surface coverage (θ_H) confirmed that the most active catalysts maintain an intermediate coverage, optimizing the availability of active sites.In conclusion, this computational screening study delivers a validated, ranked shortlist of high-performance HER catalysts. It underscores the paramount importance of tuning the hydrogen binding energy to a value near zero and provides a robust foundation for the rational design of next-generation electrocatalysts. The findings significantly contribute to the ongoing development of efficient and economically feasible technologies for green hydrogen generation, directly supporting the global transition to a sustainable energy future.

5.2 Future Scope of Work

While this computational study successfully identified promising **HER** catalysts, several avenues remain open for further investigation and development to bridge the gap between theoretical prediction and practical application.Expansion of Material Space: Future work should screen a more extensive and diverse set of materials, including ternary alloys, high-entropy alloys, single-atom catalysts on novel supports, and metal-organic frameworks (MOFs). Investigating the impact of crystal facets, defects (e.g., vacancies, grain boundaries), and nanostructuring (nanoparticles, nanowires) on the catalytic activity could reveal further performance enhancements not captured by ideal surface models.Stability and Durability Descriptors: The current screening primarily focused on activity. A critical next step is to integrate stability descriptors, such as dissolution potential, oxidation resistance, and catalyst sintering energy. A multi-objective optimization balancing both high activity and long-term durability is essential for identifying commercially viable catalysts.Advanced Microkinetic Modeling under Realistic Conditions: The model can be refined to include the effects of the electrolyte, specifically explicit solvent molecules and varying ion concentrations (pH effects). Furthermore, simulations should be extended to include a wider range of operating potentials and temperatures to predict performance under industrial electrolyzer conditions, including high current densities.